

Machine Learning – Assignment 1 Report

Algorithm Implementation

Submitted in partial fulfilment of the requirements for the award of the degree of

Ms. Data Science

IN

University of Europe for Applied Science, Dubai.

BY

Khaled Walid (70713127)

Under the guidance of

Professor Nor Azizah Hitam



2025-2026

1. Introduction (5 marks)

This assignment focuses on implementing foundational machine learning algorithms from scratch to solve a binary classification problem. The goal is to predict whether a song is popular or not based on its audio features, using the Spotify Tracks dataset. The algorithms explored include the Perceptron, Logistic Regression, and a simple Neural Network trained via backpropagation. By avoiding pre-built ML libraries for model implementation, this exercise emphasizes a deep understanding of the mathematical principles behind classification and learning processes

2. Methodology (10 marks)

Describe the approach and process followed:

- Step 1: Data collection / preparation

- The dataset was sourced from Kaggle and contains audio features for thousands of Spotify tracks.
- A binary target variable was created: songs with popularity ≥ 50 were labeled as "popular" (1), others as "not popular" (0)

- Step 2: Feature selection / preprocessing

- Selected numerical features: **danceability, energy, valence, tempo, loudness, acousticalness, and instrumentalness.**
- Features were standardized using z-score normalization to ensure consistent scale.
- The dataset was split into training and testing sets using an 80/20 ratio.

- Step 3: Model design or mathematical formulation

- **Perceptron:** Implemented using the perceptron learning rule with binary threshold activation.
- **Logistic Regression:** Used sigmoid activation and binary cross-entropy loss, optimized via gradient descent.
- **Neural Network:** Designed with one hidden layer (ReLU activation) and an output layer (sigmoid), trained using backpropagation.

- Step 4: Training and testing procedure

- Each model was trained on the training set and evaluated on the test set.
- Accuracy and RMSE (Root Mean Squared Error) were used to assess performance.
- Training iterations (`n_iter`) were varied to observe convergence behavior.

- Step 5: Tools/libraries used (e.g., NumPy, pandas, scikit-learn)

- **Python** for implementation.
- **NumPy** for matrix operations and numerical computation.
- **pandas** for data handling and preprocessing.
- **scikit-learn** only for train-test splitting and feature scaling.

3. Implementation (10 marks)

The `n_iter` parameter controls how many times the model sees the training data. For Perceptron and Logistic Regression, values between 100 and 1000 were tested. For the Neural Network, 1000 iterations were used to allow the model to learn stable weights. Reducing `n_iter` can speed up training but may lead to underfitting

4. Results and Discussion (10 marks)

Accuracy improved with higher `n_iter` values up to a point. For example, Perceptron accuracy increased from ~70% at `n_iter=100` to ~73.8% at `n_iter=1000`. Logistic Regression and Neural Network showed similar trends, plateauing around ~74.6%. This suggests that while more iterations help, gains diminish after a certain threshold.

5. Reflection (10 marks)

Implementing machine learning algorithms from scratch was both challenging and rewarding. One of the main difficulties was debugging gradient descent and ensuring proper convergence, especially in the neural network. Adjusting hyperparameters like learning rate and number of iterations required careful experimentation to balance training time and performance.

Through this process, I gained a deeper understanding of how classification models learn from data. Writing the Perceptron, Logistic Regression, and Neural Network manually helped me internalize concepts like weight updates, activation functions, and loss minimization. I also

learned how different models behave on the same dataset and how evaluation metrics like RMSE complement accuracy in assessing model confidence.

In future tasks, model performance could be improved by incorporating additional features (e.g., genre, artist), tuning hyperparameters more systematically, and using techniques like regularization or mini-batch training. Visualizing loss curves and accuracy trends across epochs would also help in diagnosing underfitting or overfitting

6. Conclusion (5 marks)

This assignment provided hands-on experience with the mathematical foundations of machine learning classification. By implementing Perceptron, Logistic Regression, and Neural Networks from scratch, I developed a stronger grasp of how models learn, optimize, and generalize. The results showed that Logistic Regression and Neural Networks achieved slightly higher accuracy (~74.6%) than the Perceptron (~73.8%), with lower RMSE values indicating better probability calibration.

Overall, this exercise reinforced the importance of preprocessing, feature selection, and iterative training in building effective ML models. These skills are essential for applying machine learning to real-world problems where interpretability and performance matter.